ANOVA 2

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Chapters for ANOVA/ANCOVA

R book

Chapter 11: Analysis of Variance and

Chapter 12: Analysis of Covariance

ISLR

Not covered per se, but you can check for relevant materials

Chapter 3, Section 3.3.1: Qualitative Predictors

Last time: two parametrizations

```r
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.sum")
round(summary(mod1)$coef,4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)     11.9     0.6241 19.0673   0.0000
## soil(clay)      -0.4     0.8826 -0.4532   0.6540
## soil(loam)       2.4     0.8826  2.7192   0.0113
```

```r
mod2 <- lm(yield~soil, data=yields, contrasts = "contr.treatment")
round(summary(mod2)$coef,4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)     11.5     1.0810 10.6385   0.0000
## soil(loam)       2.8     1.5287  1.8316   0.0781
## soil(sand)      -1.6     1.5287 -1.0466   0.3046
```

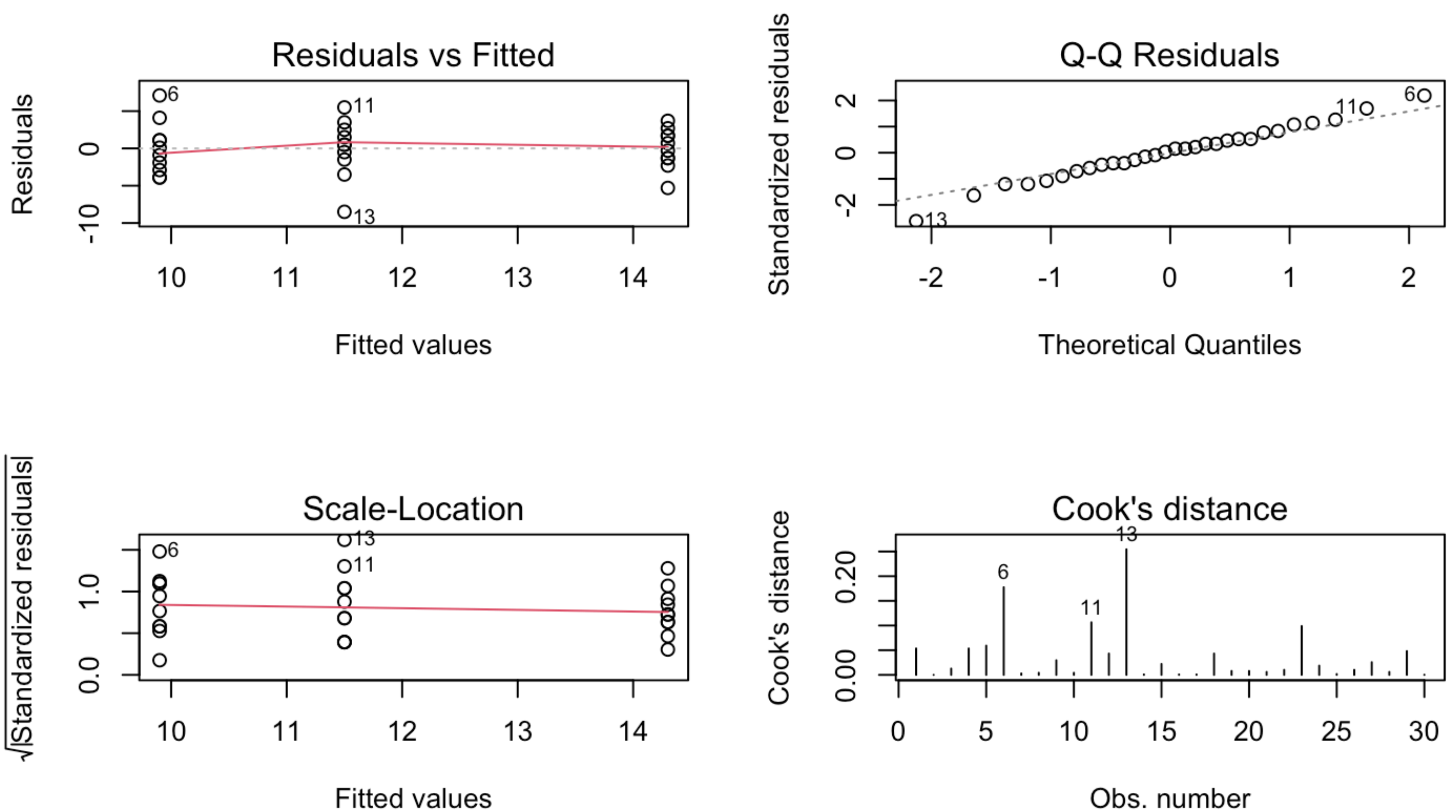
anova(mod1) and anova(mod2)

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value  Pr(>F)
## soil       2   99.2  49.600  4.2447 0.02495 *
## Residuals 27  315.5  11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value  Pr(>F)
## soil       2   99.2  49.600  4.2447 0.02495 *
## Residuals 27  315.5  11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual analysis

We should also for ANOVA-models check the assumptions by residual analysis



ANOVA with several factors

- Each with two or more levels
- Assume two factors: A with $i = 1, 2, \ldots a$ levels, and B with $j = 1, 2, \ldots, b$ levels
- If all levels of A are tested with all levels of B we have a complete crossed design
- Assume there are $k = 1, 2, \ldots, r$ replicates of all ab level combinations so
- So, the total number of observations is $n = a \cdot b \cdot r$

Illustration of two-factor case

https://solve.shinyapps.io/Factorialdemo/

The Model

Model:

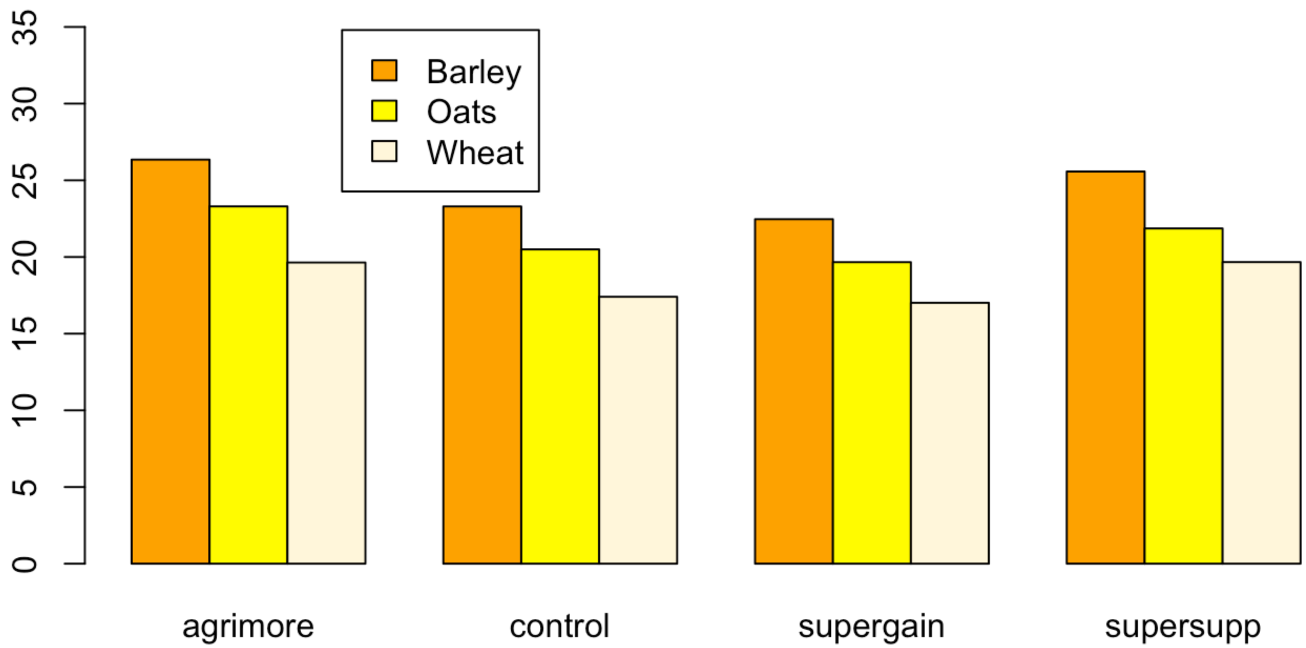
$$y_{ijk} = \mu_{ij} + \epsilon_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where $\epsilon_{ijk} \sim N(0, \sigma^2)$ and $(\alpha\beta)_{ij}$ is the effect due to any interaction between the *i*-th level of A and the *j*-th level of B.

Parametrization (restrictions)

- Sum-to-zero: $\sum_i \alpha_i = 0$, $\sum_j \beta_j = 0$, $\sum_i (\alpha\beta)_{ij} = 0$ and $\sum_j (\alpha\beta)_{ij} = 0$.
- Reference levels: $\alpha_1 = 0$, $\beta_1 = 0$, $(\alpha\beta)_{1j} = 0$ for all j and $(\alpha\beta)_{i1} = 0$ for all i.

Example: Animal diets - “growth data” from R book



R structure of the data

```
##    supplement  diet     gain
## 1   supergain wheat 17.37125
## 2   supergain wheat 16.81489
## 3   supergain wheat 18.08184
## 4   supergain wheat 15.78175
## 5     control wheat 17.70656
## 6     control wheat 18.22717
## 7     control wheat 16.08650
## 8     control wheat 17.60184
## 9   supersupp wheat 20.78861
## 10  supersupp wheat 20.00858
## 11  supersupp wheat 19.30287
## 12  supersupp wheat 18.57330
## 13   agrimore wheat 19.54688
## 14   agrimore wheat 17.96440
## 15   agrimore wheat 21.43738
## 16   agrimore wheat 19.60763
```

R analysis

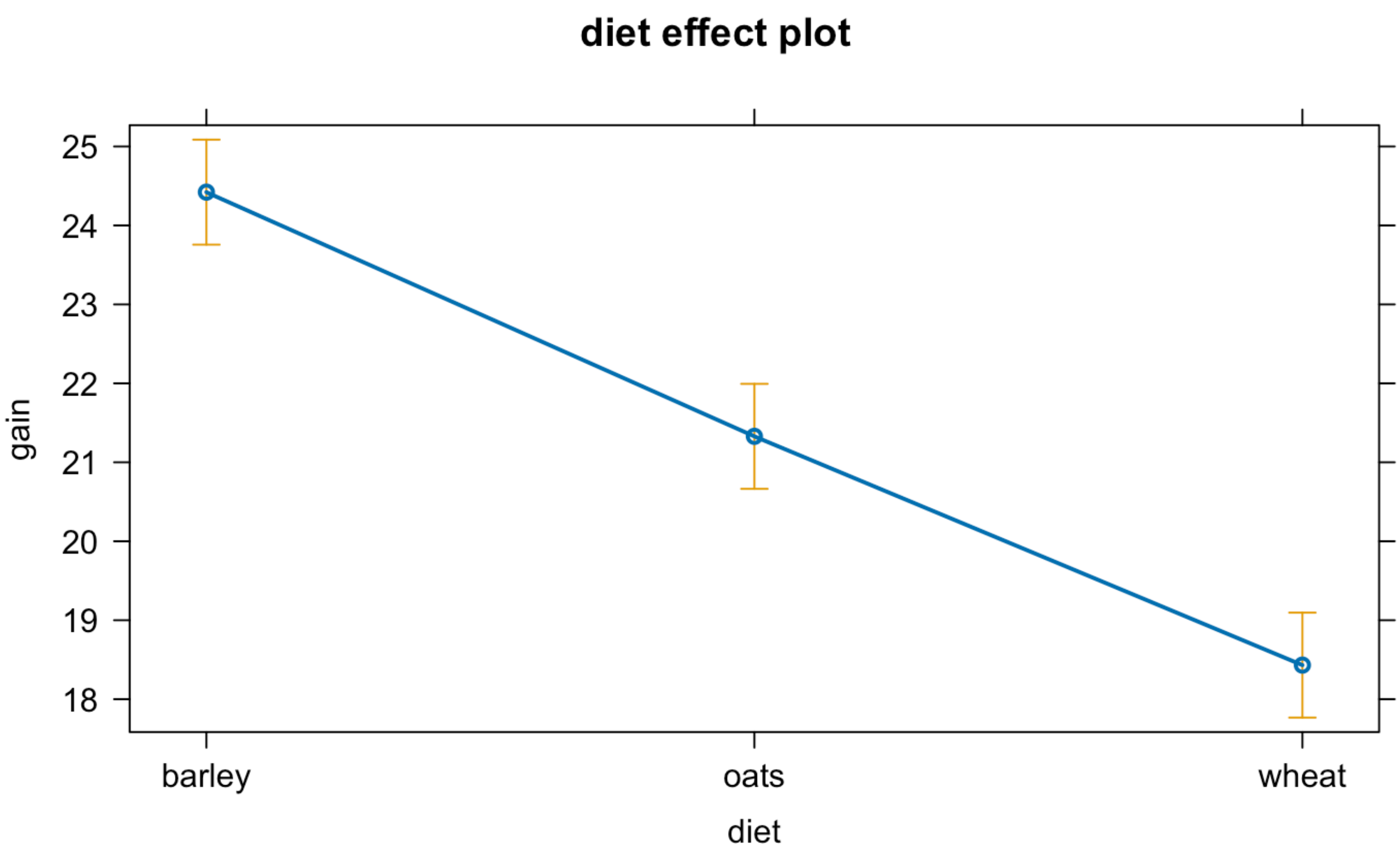
Let's see how to use both parametrizations here:

```
mod2 <- mixlm::lm(gain ~ diet + supplement + diet:supplement,
           data=weights, contrasts = "contr.sum")

options(contrasts = c("contr.sum", "contr.poly"))
mod2 <- stats:::lm(gain ~ diet + supplement + diet:supplement,
           data=weights)

mod1<- mixlm::lm(gain ~ diet + supplement + diet:supplement,
           data=weights, contrasts = "contr.treatment")

options(contrasts = c("contr.treatment", "contr.poly"))
mod1 <- stats:::lm(gain ~ diet + supplement + diet:supplement,
           data=weights)
```

Main effects plot of `Diet`

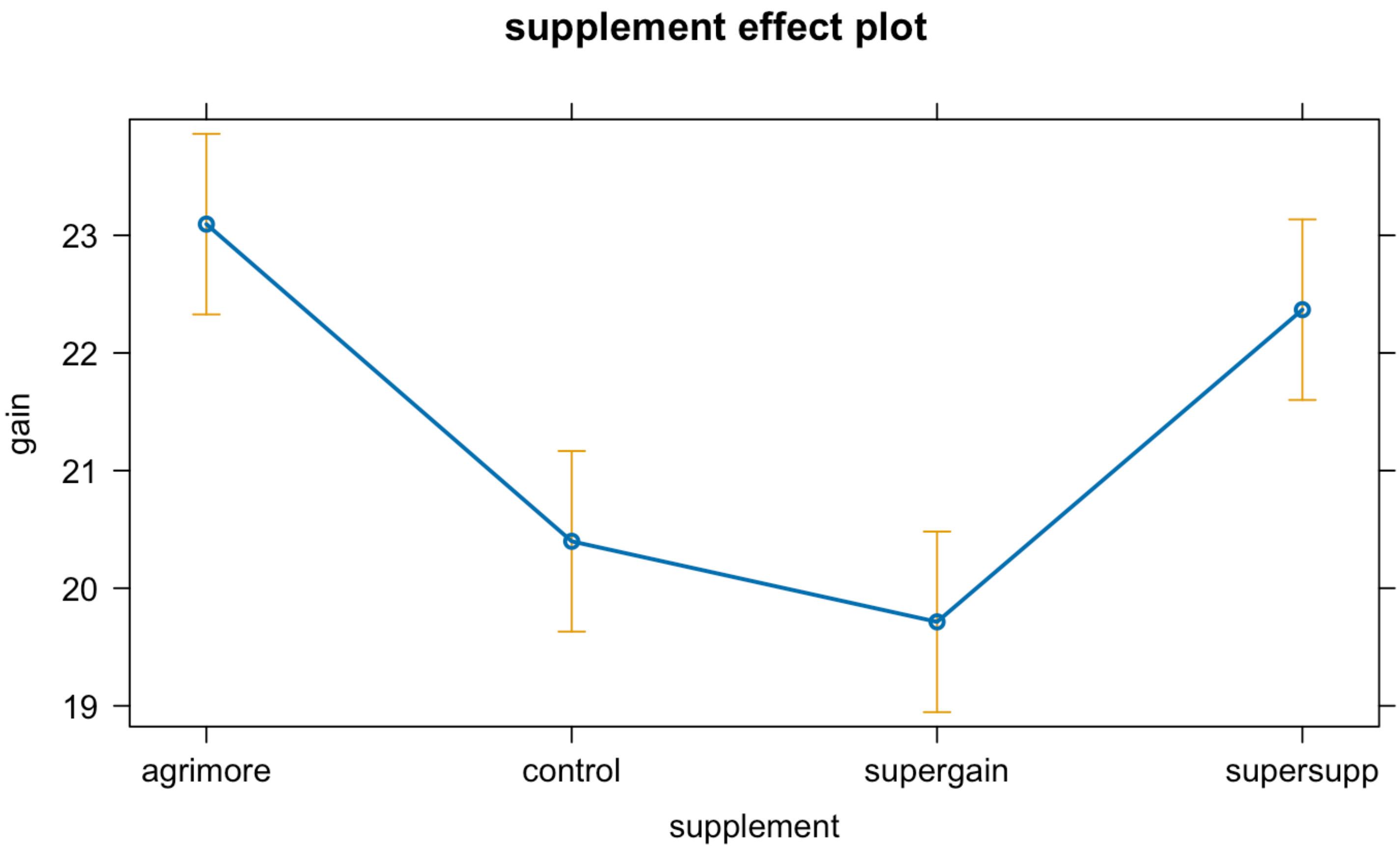
"Main effect plots" of group means with 95% CI's

```
library(effects)
plot(Effect(focal.predictors = c("diet"), mod = mod2))
```



Main effects plot of `Supplement`

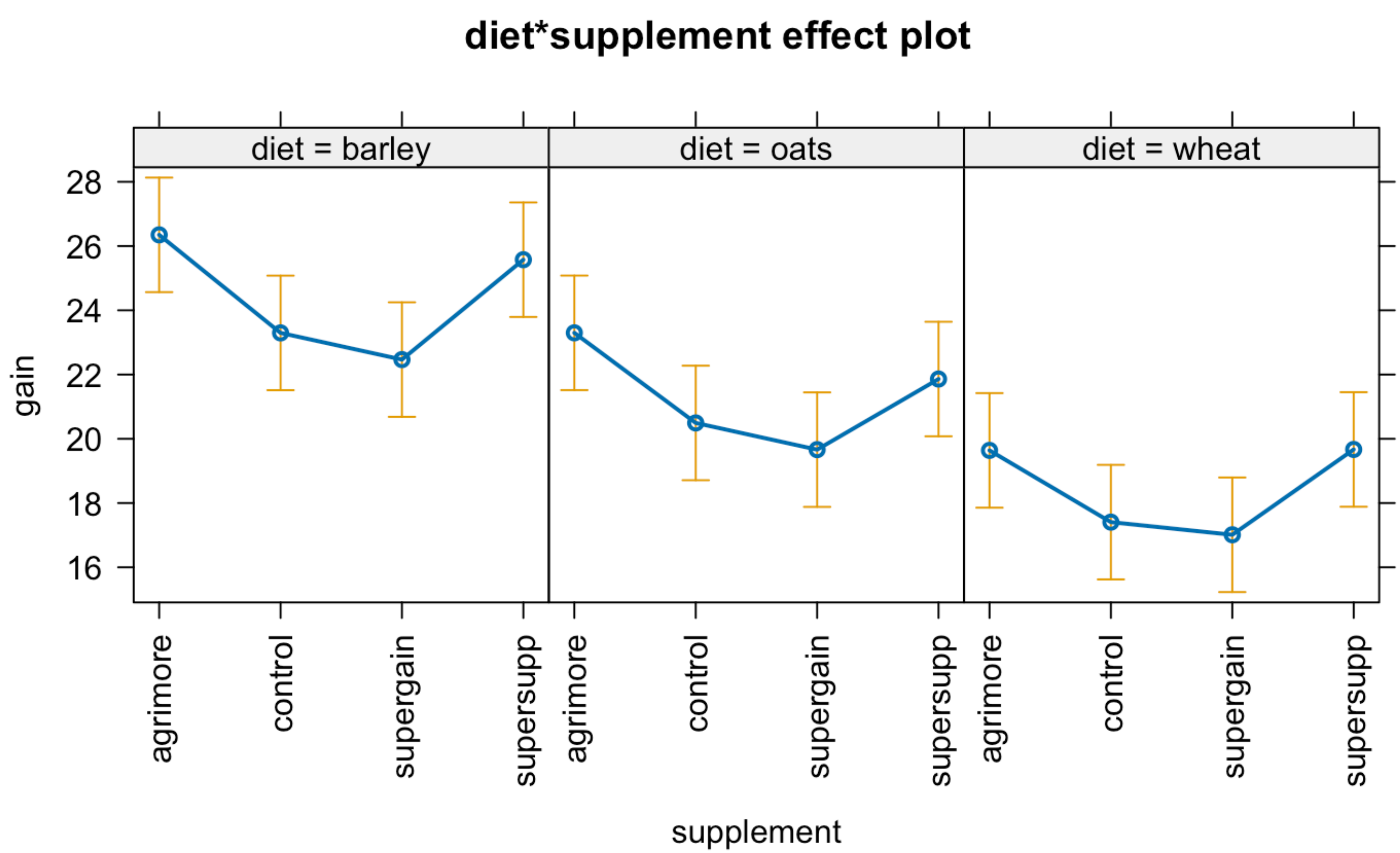
```
plot(Effect(c("supplement"), mod2))
```



Interaction plots

Means of diet:supplement combinations.

```
plot(allEffects(mod2,confidence.level=.99),layout=c(3,1),
     rotx=90,alternating=FALSE)
```



Model parameter estimation

Still likelihood maximization/ SSE minimization in the regression formulation of 2-way ANOVA with interactions to get $\widehat{\beta_0}, \widehat{\beta_1}, \ldots, \widehat{\beta_p}$ which can similarly to how we did it last time translated into $\widehat{\mu}, \widehat{\alpha}_i, \widehat{\beta}_j, \widehat{(\alpha\beta)}_{ij}$. For balanced data the least squares estimators for the model parameters are (sum-to-zero):

$$\widehat{\mu} = \bar{y}_{...}$$

$$\widehat{\alpha}_i = \bar{y}_{i..} - \bar{y}_{...}$$

$$\widehat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}$$

$$\widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$$

Interpretation: effects are **deviations** from the overall mean μ.

Model parameter estimation

Now take level 1 of A and B as reference

$$\widehat{\mu}^{(ref)} = \bar{y}_{11.}$$

$$\hat{\alpha}_i^{(ref)} = \bar{y}_{i1.} - \bar{y}_{11.}, \quad i = 2, \ldots, a$$

$$\hat{\beta}_j^{(ref)} = \bar{y}_{1j.} - \bar{y}_{11.}, \quad j = 2, \ldots, b$$

$$\widehat{(\alpha\beta)}_{ij}^{(ref)} = \bar{y}_{ij.} - \bar{y}_{i1.} - \bar{y}_{1j.} + \bar{y}_{11.}, \quad i = 2, \ldots, a, \; j = 2, \ldots, b$$

Interpretation: effects are **differences** from the reference.

As usual, the noise variance estimate is:

$$\hat{\sigma}^2 = MSE$$

Summary with sum to zero:

Sum to zero parametrization - deviations from the grand mean:

```
##                                     Estimate Std. Error  t value Pr(>|t|)
## (Intercept)                          21.3939     0.1893 113.0450   0.0000
## diet(barley)                          3.0277     0.2676  11.3125   0.0000
## diet(oats)                           -0.0651     0.2676  -0.2433   0.8092
## supplement(agrimore)                  1.7014     0.3278   5.1904   0.0000
## supplement(control)                  -0.9953     0.3278  -3.0364   0.0044
## supplement(supergain)                -1.6801     0.3278  -5.1254   0.0000
## diet(barley):supplement(agrimore)     0.2255     0.4636   0.4864   0.6297
## diet(oats):supplement(agrimore)       0.2682     0.4636   0.5785   0.5665
## diet(barley):supplement(control)     -0.1297     0.4636  -0.2797   0.7813
## diet(oats):supplement(control)        0.1602     0.4636   0.3455   0.7317
## diet(barley):supplement(supergain)   -0.2754     0.4636  -0.5942   0.5561
## diet(oats):supplement(supergain)      0.0143     0.4636   0.0308   0.9756
```

Summary with reference:

Reference diet: **barley**, supplement: **agrimore**

```
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       26.3485     0.6556 40.1907   0.0000
## diet(oats)                        -3.0501     0.9271 -3.2898   0.0022
## diet(wheat)                       -6.7094     0.9271 -7.2367   0.0000
## supplement(control)               -3.0518     0.9271 -3.2917   0.0022
## supplement(supergain)             -3.8824     0.9271 -4.1875   0.0002
## supplement(supersupp)             -0.7732     0.9271 -0.8339   0.4098
## diet(oats):supplement(control)     0.2471     1.3112  0.1885   0.8516
## diet(wheat):supplement(control)    0.8183     1.3112  0.6241   0.5365
## diet(oats):supplement(supergain)   0.2470     1.3112  0.1884   0.8517
## diet(wheat):supplement(supergain)  1.2557     1.3112  0.9577   0.3446
## diet(oats):supplement(supersupp)  -0.6650     1.3112 -0.5072   0.6151
## diet(wheat):supplement(supersupp)  0.8024     1.3112  0.6120   0.5444
```

We want to test hypotheses

Interaction effects?

$$H_0 : (\alpha\beta)_{11} = (\alpha\beta)_{12} = \cdots = (\alpha\beta)_{ab} = 0$$

Main effect of A?

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$$

Main effect of B?

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = 0$$

Splitting of sums of squares

The Total Sums of Squares measures total variability:

$$SST = \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{...})^2$$

Can be split into:

$$SST = SSA + SSB + SSAB + SSE$$

Degrees of freedom follow correspondingly as

$$(n - 1) = [(a - 1)] + [(b - 1)] + [(a - 1)(b - 1)] + [n - ab].$$

This corresponds to a linear regression formulation with $SSR = SSA + SSB + SSAB$ and $p = ab - 1$

F-tests

Interaction: Reject H_0 if

$$F = \frac{MSAB}{MSE} = \frac{SSAB/[(a-1)(b-1)]}{SSE/[n-ab]}$$

is larger than $F_{\alpha,\,(a-1)(b-1),\,n-ab}$ or p-value < α.

Main effect A: Reject H_0 if

$$F = \frac{MSA}{MSE} > F_{\alpha,\,(a-1),\,n-ab}$$

Main effect B: Reject H_0 if

$$F = \frac{MSB}{MSE} > F_{\alpha,\,(b-1),\,n-ab}$$

The ANOVA-table for the growth data

```
anova(mod2)
```

```
## Analysis of Variance Table
##
## Response: gain
##                 Df  Sum Sq Mean Sq F value    Pr(>F)
## diet             2 287.171 143.586 83.5201 2.999e-14 ***
## supplement       3  91.881  30.627 17.8150 2.952e-07 ***
## diet:supplement  6   3.406   0.568  0.3302    0.9166
## Residuals       36  61.890   1.719
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions?

A reduced model

```
mod3 <- lm(gain ~ diet + supplement, data=weights, contrasts = "contr.sum")
anova(mod3)
```

```
## Analysis of Variance Table
##
## Response: gain
##            Df  Sum Sq Mean Sq F value    Pr(>F)
## diet        2 287.171 143.586  92.358 4.199e-16 ***
## supplement  3  91.881  30.627  19.700 3.980e-08 ***
## Residuals  42  65.296   1.555
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: The complete crossed design gives “orthogonal effects”. The SS’s are independent, and, apart from SSE, they do not change when effects are removed/added to the model.

Tukey tests supplement

```
ptests <- mixlm::simple.glht(mod3, "supplement", corr = c("Tukey"),
                             level = 0.95)
print(round(ptests$res,4))
```

```
##                       Lower  Center   Upper Std.Err t value  P(>t)
## agrimore-control     1.3351  2.6967  4.0583   0.509  5.2977 0.0000
## agrimore-supergain   2.0198  3.3815  4.7431   0.509  6.6429 0.0000
## agrimore-supersupp  -0.6343  0.7274  2.0890   0.509  1.4289 0.4889
## control-supergain   -0.6769  0.6848  2.0464   0.509  1.3452 0.5400
## control-supersupp   -3.3310 -1.9693 -0.6077   0.509 -3.8688 0.0020
## supergain-supersupp -4.0157 -2.6541 -1.2925   0.509 -5.2141 0.0000
```

Tukey tests diet

```
ptests2 <- mixlm::simple.glht(mod3, "diet", corr = c("Tukey"),
                              level = 0.95)
print(round(ptests2$res,4))
```

```
##                Lower Center  Upper Std.Err t value P(>t)
## barley-oats  2.0218 3.0928 4.1638  0.4408  7.0159     0
## barley-wheat 4.9193 5.9903 7.0613  0.4408 13.5886     0
## oats-wheat   1.8265 2.8975 3.9685  0.4408  6.5727     0
```

Compact letter display

A compact letter display (cld) of groups of effects

```
cld(ptests)
```

```
## Tukey's HSD
## Alpha: 0.05
##
##               Mean G1 G2
## agrimore  23.09531     B
## supersupp 22.36796     B
## control   20.39861  A
## supergain 19.71385  A
```

Analysis of covariance - ANCOVA

The dataset `birth` in the BIAS.data package holds records of 189 birth weights from Massachusetts, USA, and some additional variables. Three of the variables are,

Var	Description
`LWT`	Weight of mother.
`SMK`	`YES` if mother smokes and `NO` otherwise.
`BWT`	Birth weight (RESPONSE).

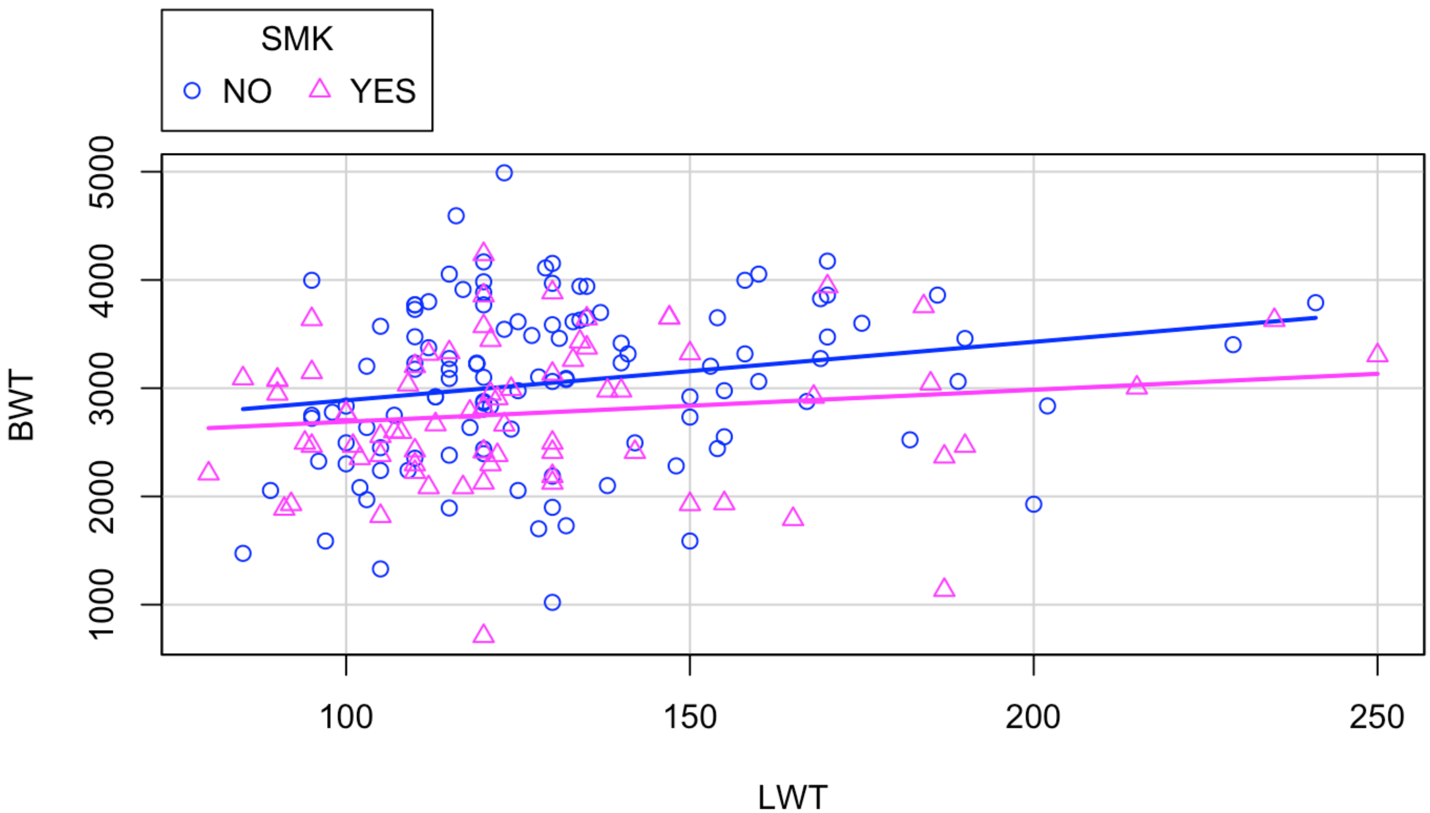
The main question is: Does mother's smoking affect birth weight and does it depend on the mother weight?

We have both a categorical and a continuous predictor for the response variable.

The data

```
##   LWT SMK  BWT
## 1 120 YES  709
## 2 130  NO 1021
## 3 187 YES 1135
## 4 105  NO 1330
## 5  85  NO 1474
## 6 150  NO 1588
## 7  97  NO 1588
```

Scatterplot with separate linear models for smokers and non-smokers



ANCOVA as regression with indicator variable

Let $y = BWT$ and $x_1 = LWT$ and define

$$x_{2i} = \begin{cases} 0 & \text{if smoker} \\ 1 & \text{if non-smoker} \end{cases}$$

Assume the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$$

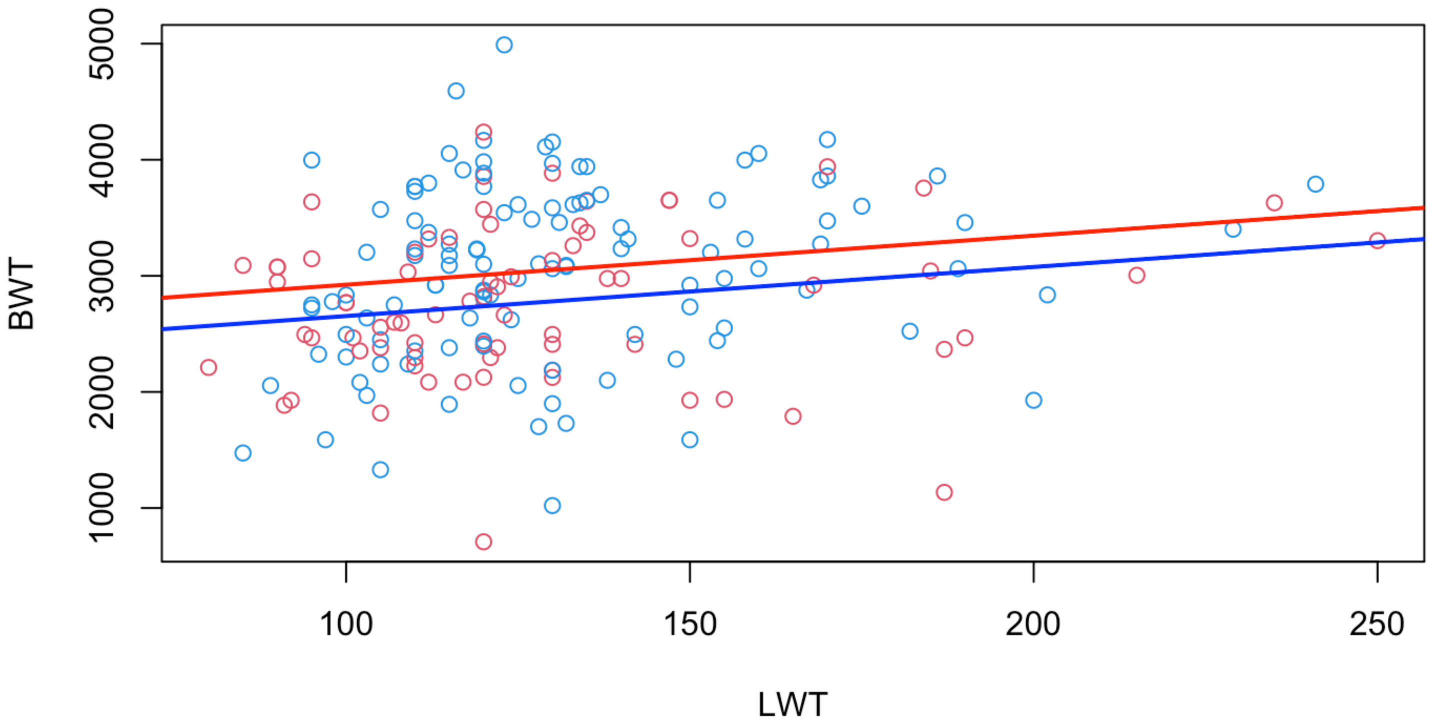
where $\epsilon_i \sim N(0, \sigma^2)$.

Fitting the model in R

```
birth4 <- lm(BWT ~ LWT + SMK, data=birth, contrasts = "contr.treatment")
summary(birth4)
```

```
##
## Call:
## lm(formula = BWT ~ LWT + SMK, data = birth, contrasts = "contr.treatment")
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2030.16  -447.00    27.74   514.20  1968.51
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2500.174    230.833  10.831   <2e-16 ***
## LWT            4.238      1.690   2.508   0.0130 *
## SMK(YES)    -270.013    105.590  -2.557   0.0114 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## s: 707.8 on 186 degrees of freedom
## Multiple R-squared: 0.06731,
## Adjusted R-squared: 0.05728
```

The estimated model(s)



The model with interaction term

Assume that we expand the model with an interaction term between LWT and SMK

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} \cdot x_{2i} + \epsilon_i$$

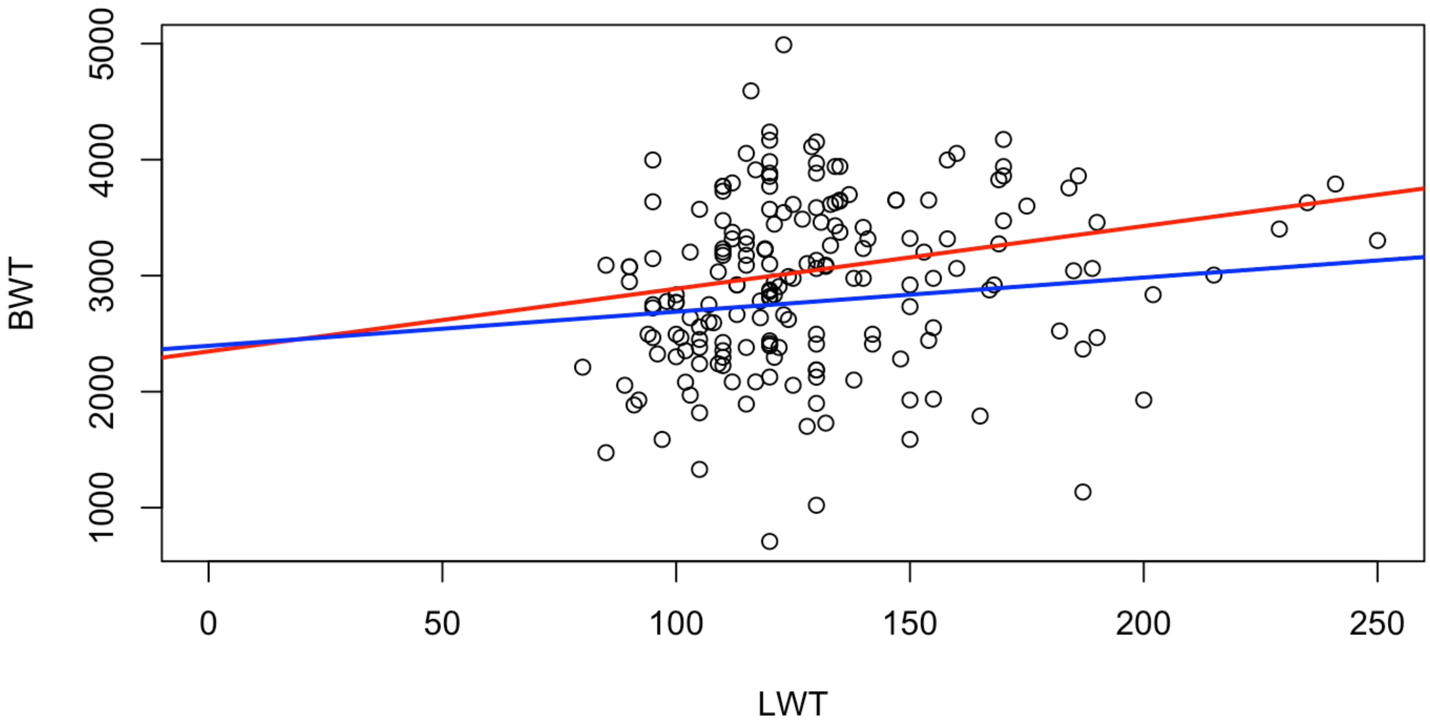
with $\epsilon_i \sim N(0, \sigma^2)$.

Lets see how the model is for different values of x_{2i}

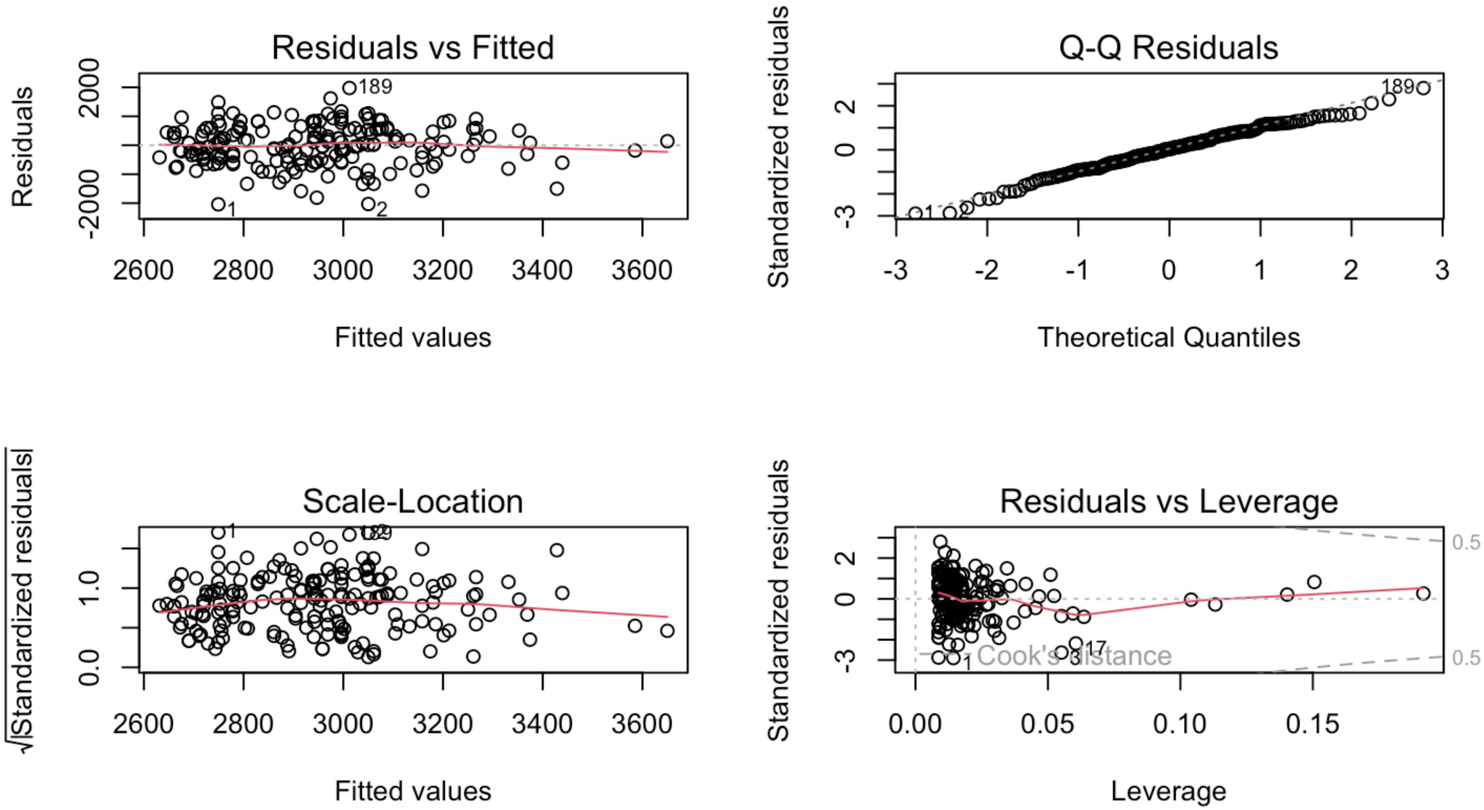
Fitted model in R

```
##
## Call:
## lm(formula = BWT ~ LWT * SMK, data = birth, contrasts = "contr.treatment")
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2040.25  -456.20    29.07   531.72  1977.72
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2347.507    312.717   7.507 2.48e-12 ***
## LWT             5.405      2.335   2.315   0.0217 *
## SMK(YES)       47.867    451.163   0.106   0.9156
## LWT:SMK(YES)   -2.456      3.388  -0.725   0.4695
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## s: 708.7 on 185 degrees of freedom
## Multiple R-squared: 0.06995,
## Adjusted R-squared: 0.05487
## F-statistic: 4.638 on 3 and 185 DF,  p-value: 0.003756
```

The estimated model(s)



Model summaries, tests, diagnostics



Model summaries, tests, diagnostics

You should be able to check:

- Are model assumptions met?
- Are there significant effects of LWT and/or SMK?
- How is the model fit in terms of R^2? Critizise the model!

Assignment posted today!